Notes: Sharifkhani & Simutin (JFE, 2021)

Feedback loops in industry trade networks and the term structure of momentum profits

Contents

1 Big picture

- **Question.** Why do momentum profits come mostly from intermediate-horizon returns rather than the most recent past returns, and what economic mechanism generates this "momentum echo"?
- **Puzzle.** Novy-Marx (2012) shows that the strongest momentum profits are based on returns from about seven to twelve months ago, not from returns in the immediately preceding months. Standard momentum models usually predict that predictive power should decay with lag.
- **Main mechanism.** Industries are connected through customer-supplier trade links. A shock to one industry can travel through the industry trade network and later feed back to the original industry. This delayed feedback creates intermediate-horizon autocorrelation in industry returns.
- **Key object.** The paper constructs a feedback-strength measure, FS, which captures how strongly an industry is connected back to itself through intersectoral trade loops.
- **Main empirical finding.** Momentum echo profits are much larger in industries with high feedback strength. The intermediate-horizon winner-minus-loser strategy earns about 0.95% per month in high-FS industries but only about 0.38% per month in low-FS industries.
- **Recent returns do not drive the result.** Sorts based on recent-horizon returns, measured using returns from months $t-6$ to $t-2$, do not generate the same positive relation with feedback strength.
- **Mechanism evidence.** The relation between feedback strength and echo profits is stronger when information diffusion is more limited along the feedback loop, proxied by lower analyst coverage across the industries in the loop.
- **Direct feedback-loop evidence.** Returns of industries located on the strongest feedback loop help explain whether a shock returns to the original industry. The echo effect is stronger when returns along the actual feedback-loop portfolio move consistently with the original shock.
- **Robustness.** Results survive alternative feedback-strength measures, controls for industry centrality and size, and various modifications to the construction of the network.
- **Economic takeaway.** Industry trade networks are not only conduits for cross-industry spillovers. Their loop structure also determines the timing of return predictability within an industry.

2 Data and measurement

2.1 Industry return data

- The return sample covers January 1973 through December 2016.
- The paper studies 49 industry portfolios.
- Industry portfolio returns and market capitalizations are obtained from CRSP and Ken French's data library.
- Portfolios are value weighted.
- The analysis focuses on industry-level returns rather than firm-level returns because the hypothesized propagation channel is intersectoral trade.

Interpretation: The industry level is the natural unit because the mechanism concerns shocks moving through trade links among industries. A firm-level shock may travel through suppliers and customers, but the feedback-loop measure is built from aggregate input-output relationships.

2.2 Input-output network data

- The paper constructs an intersectoral trade network using Bureau of Economic Analysis Input-Output Accounts.
- The network spans 1972–2015.
- Nodes are industries.
- Directed links represent customer-supplier trade flows.
- Link weights capture the economic importance of one industry to another.

Interpretation: The BEA input-output data provide an empirical map of how industries buy from and sell to one another. This map determines how industry shocks can propagate.

2.3 Customer and supplier influence weights

For industries i and j, the paper constructs two directed influence measures.

- w_{ij}^C measures the importance of industry i to industry j as a customer.
- w_{ij}^S measures the importance of industry i to industry j as a supplier.
- The overall influence weight is

$$\widetilde{w}_{ij} = \frac{w_{ij}^C + w_{ij}^S}{2}. \tag{1}$$

Interpretation: Using both customer and supplier measures makes the trade network directional but symmetric in economic interpretation: a shock can propagate because one industry is an important buyer, an important supplier, or both.

2.4 Feedback strength

The feedback-strength measure is designed to identify the strongest route through which a shock can leave an industry and come back to the same industry.

- The paper interprets the inverse of the influence weight as a distance:

$$d_{ij} = \widetilde{w}_{ij}^{-1}. \tag{2}$$

- It applies Dijkstra's algorithm to identify the strongest feedback path from an industry back to itself.
- To prevent the algorithm from selecting the trivial zero-length path from an industry to itself, the paper creates a "shadow self" node that inherits inbound but not outbound links.
- Feedback strength is high when the loop back to the industry contains strong trade links.
- An isolated industry has FS = 0.

Interpretation: FS is a loop-strength measure, not simply a centrality measure. An industry can be central in the overall network without having a strong path that feeds shocks back to itself.

2.5 Intermediate-horizon and recent-horizon returns

The paper distinguishes between two lookback windows.

$$\mathrm{IR}_{i,t} = \sum_{k=7}^{11} r_{i,t-k}, \tag{3}$$

$$\mathrm{RR}_{i,t} = \sum_{k=2}^{6} r_{i,t-k}. \tag{4}$$

- IR is the intermediate-horizon return, using returns from months $t-11$ through $t-7$.
- RR is the recent-horizon return, using returns from months $t-6$ through $t-2$.
- The echo strategy primarily uses IR because the momentum echo is an intermediate-horizon phenomenon.

Interpretation: The distinction between IR and RR is the key empirical signature. If feedback loops generate an echo, the effect should show up in IR-based momentum but not necessarily in recent-return momentum.

2.6 Analyst coverage along feedback loops

- The paper uses I/B/E/S analyst coverage to proxy for information diffusion frictions.
- For each pair of successive industries on a feedback loop, the paper counts analysts covering firms in both industries.
- The count is normalized by the total number of firms in the outbound industry.
- The loop-level measure is the geometric average of pairwise analyst coverage along the strongest feedback loop.

Interpretation: If analysts understand multiple industries on the loop, they can transmit information more quickly. The echo should therefore be stronger when analyst coverage along the loop is lower.

3 Empirical design and specifications

3.1 Baseline echo portfolio sort

At the end of month $t-1$:

1. Sort industries into terciles by feedback strength FS.
2. Within each FS tercile, sort industries into quartiles by intermediate-horizon returns IR.
3. Form a winner-minus-loser portfolio in month t by buying the top IR quartile and shorting the bottom IR quartile.
4. Hold the value-weighted portfolio for one month.

Interpretation: This design asks whether the profitability of intermediate-horizon momentum depends on how strongly an industry is connected back to itself through trade loops.

3.2 Risk adjustment

Portfolio returns are evaluated using:

- raw excess returns;
- CAPM alphas;
- Fama-French three-factor alphas;
- Fama-French four-factor alphas.

A generic alpha regression is

$$R_{p,t} - R_{f,t} = \alpha_p + \beta_p' F_t + \epsilon_{p,t}, \tag{5}$$

where F_t denotes the selected factor vector.

Interpretation: Showing that echo profits survive factor adjustment rules out a simple explanation based on standard risk exposures.

3.3 Term-structure analysis

The term-structure analysis constructs winner-minus-loser portfolios using returns measured at lag k.

- In the one-month formation version, industries are sorted by $r_{i,t-k}$.
- In the five-month formation version, industries are sorted by cumulative returns from $t-k$ through $t-k-4$.
- The strategy is evaluated for lags $k = 1, \ldots, 12$.

Interpretation: The model predicts a wave-like pattern for high-FS industries: weak recent predictability, stronger intermediate-lag predictability, and then fading returns.

3.4 Conditional echo strategy

The conditional echo strategy further conditions on the formation-period feedback strength of winner industries.

- The winner portfolio is formed using intermediate-horizon returns.
- The strategy is implemented only when the feedback strength of the winner portfolio is in the top quartile or top decile of its time-series distribution.
- The idea is that echo profits should be strongest when the industries being traded have unusually strong feedback loops during the relevant formation period.

Interpretation: This test adds time-series variation. It asks not only whether high-FS industries have larger echo profits, but whether the echo strategy is especially profitable when feedback strength is high at the moment the strategy is formed.

3.5 Limited-information regression

The paper estimates panel regressions of industry factor-adjusted returns on prior returns, feedback strength, analyst coverage, and interactions. A simplified version is

$$r_{i,t}^{adj} = a + b_1 r_{i,t-1} + b_2 \mathrm{IR}_{i,t} + b_3 \mathrm{RR}_{i,t} + b_4 \mathrm{FS}_{i,t} + b_5 (\mathrm{FS}_{i,t} \times \mathrm{IR}_{i,t}) + b_6 Coverage_{i,t} + b_7 (Coverage_{i,t} \times \mathrm{FS}_{i,t} \times \mathrm{IR}_{i,t}) + u_{i,t}. \tag{6}$$

Interpretation: The core prediction is $b_5 > 0$ and $b_7 < 0$: feedback strength amplifies intermediate-horizon predictability, and the amplification is weaker when analyst coverage along the loop is high.

3.6 Feedback-loop portfolio regression

To trace the shock through the loop, the paper constructs a feedback-loop portfolio, FLP, for each industry.

- FLP consists of industries on the strongest feedback loop that are most influenced by the original industry.
- The paper interacts FLP returns with FS and IR.
- A placebo version uses industries that are similarly influenced by the original industry but are not on the strongest feedback loops.

A simplified regression is

$$r_{i,t}^{adj} = a + b_1 r_{i,t-1} + b_2 \mathrm{IR}_{i,t} + b_3 \mathrm{FLP}_{i,t} + b_4 \mathrm{FS}_{i,t} + b_5 (\mathrm{FS}_{i,t} \times \mathrm{IR}_{i,t} \times \mathrm{FLP}_{i,t}) + u_{i,t}. \quad (7)$$

Interpretation: This test asks whether the echo is strongest when returns of industries along the actual feedback loop align with the shock that is expected to feed back.

4 Identification and interpretation

4.1 What identifies the mechanism

- The mechanism is identified by cross-sectional differences in feedback strength across industries and time variation in industry returns.
- High-FS and low-FS industries are exposed to the same factor models and broad market shocks, but differ in the structure of their trade loops.
- The key tests focus on intermediate-horizon returns, which is where the echo mechanism predicts delayed feedback should appear.
- Mechanism tests use analyst coverage and feedback-loop portfolio returns to test whether shocks actually diffuse slowly along the trade network.

Interpretation: The paper does not rely on a single return sort. It links the return pattern to network topology, information frictions, and returns along the actual feedback loop.

4.2 Why centrality alone is not enough

- An industry can be central in the input-output network without having a strong loop back to itself.
- Centrality and industry size are determinants of feedback strength, but they do not fully explain the echo effect.
- Robustness tests control for centrality and the number of firms in the industry.

Interpretation: The economic object is a feedback loop, not merely an industry's importance in the aggregate production network.

4.3 Why recent returns are a placebo

- A general momentum explanation would predict recent-horizon returns to matter as well.
- The feedback-loop hypothesis predicts strongest effects at intermediate horizons, after shocks have propagated through trade links and returned.

- The weak relation between FS and recent-return momentum supports the feedback-loop interpretation.

Interpretation: The term structure of predictability is the key evidence. The model predicts not just momentum, but momentum at the echo horizon.

4.4 Limitations

- The paper measures trade links at the industry level, not at the firm-pair level.
- Input-output data may not perfectly capture financial-market attention or information flows.
- The feedback-strength measure depends on graph-theoretic choices, such as using Dijkstra's algorithm and focusing on the strongest loop.
- The mechanism is tested indirectly through return predictability, analyst coverage, and loop-portfolio returns rather than direct observation of investor beliefs.

Interpretation: The paper's strength is that several different empirical designs point to the same mechanism: delayed feedback of shocks through industry trade loops.

5 Tables and figures

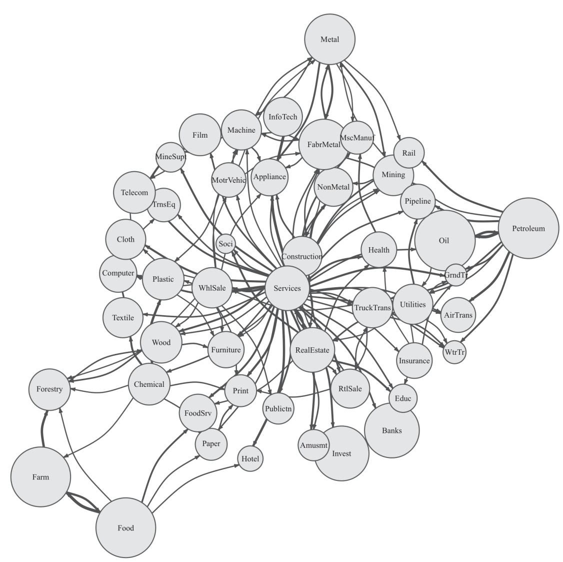
Visuals are directly extracted from the paper and grouped compactly to save space.

5.1 Figure 1 and Table 1: trade network and feedback-strength distribution

Read:

- Figure 1 shows the customer-supplier trade network among 49 industries in 2015.
- Node size represents industry size, and edge thickness represents connection strength.
- Table 1 ranks industries by average feedback strength.
- The highest-FS industries are petroleum and coal products, oil and gas extraction, farms, and food and beverage products.
- The lowest-FS industries include social assistance, transit and ground passenger transportation, water transportation, and accommodation.

Economic message: Feedback strength is not evenly distributed. Some industries sit in dense trade loops, while others have weak or almost nonexistent paths back to themselves.



ig. 1. Trade network of 49 industries in 2015.
his figure plots the customer-supplier trade network of the 49 industries constructed based on the Bureau of Economic Analysis Input-Output Accounts ı 2015. The size of the circles (nodes) represents the size of the industry, and the thickness of lines captures the magnitude (weight) of trade connections etween industry pairs. Industry keys in the middle of the circles correspond to the keys in the last column of Table 1. The sample is described in ppendix A. The calculation of industry connections is described in Section 2.1. To avoid crowding the figure, only connections with the weights of 5% or

Figure 1: trade network of 49 industries

Table 1
Industry feedback strength summary statistics.
This table reports summary statistics for the 49 industries in the sample, ranked by their average feedback strength (FS). Each month, an industry's FS is calculated by applying Dijkstra's algorithm on the matrix of direct intersectoral supply chain connections as described in Section 2.1. The table reports the time-series mean, standard deviation, minimum, and maximum of FS for each industry from 1973 to 2015. The sample is described in Appendix A. The last column shows the industry keys used to label industries in Fig. 1.

		Feedback strength				
Rank	Industry	Mean	St. dev	Min	Max	Industry key
1	Petroleum and coal products	0.135	0.014	0.117	0.160	Petroleum
2	Oil and gas extraction	0.135	0.014	0.117	0.160	Oil
3	Farms	0.119	0.007	0.106	0.128	Farm
4	Food and beverage and tobacco products	0.119	0.007	0.106	0.128	Food
5	Fabricated metal products	0.074	0.004	0.063	0.078	FabrMetal
6	Primary metals	0.074	0.004	0.063	0.078	Metal
7	Funds, trusts, Fed. banks, credit intermediation	0.063	0.023	0.034	0.100	Banks
8	Securities, commodity contracts, and investments	0.062	0.025	0.025	0.100	Invest
9	Textile mills and textile product mills	0.061	0.020	0.015	0.082	Textile
10	Apparel and leather and allied products	0.061	0.020	0.015	0.082	Cloth
11	Utilities	0.056	0.015	0.035	0.080	Utilities
12	Wood products	0.055	0.009	0.037	0.070	Wood
13	Forestry, fishing, and related activities	0.055	0.009	0.037	0.070	Forestry
14	Mining, except oil and gas	0.051	0.018	0.025	0.080	Mining
15	Services	0.047	0.006	0.039	0.058	Services
16	Real estate	0.044	0.008	0.033	0.058	RealEstate
17	Chemical products	0.044	0.005	0.037	0.053	Chemical
18	Plastics and rubber products	0.044	0.005	0.037	0.053	Plastic
19	Pipeline transportation	0.043	0.018	0.018	0.069	Pipeline
20	Construction	0.042	0.005	0.033	0.051	Construction
21	Food services and drinking places	0.042	0.010	0.026	0.052	FoodSrv
22	Nonmetallic mineral products	0.039	0.005	0.031	0.048	NonMetal
23	Machinery	0.039	0.007	0.029	0.047	Machine
24	Printing and related support activities	0.038	0.010	0.016	0.051	Print
25	Wholesale trade	0.037	0.005	0.032	0.048	WhlSale
26	Paper products	0.037	0.011	0.013	0.051	Paper
27	Broadcasting and telecommunications	0.036	0.006	0.029	0.046	Telecom
28	Motion picture and sound recording industries	0.036	0.006	0.029	0.046	Film
29	Retail trade	0.032	0.005	0.022	0.039	RtlSale
30	Publishing industries, except internet (includes software)	0.031	0.010	0.015	0.043	Publish
31	Furniture and related products	0.028	0.003	0.023	0.037	Furniture
32	Motor vehicles, bodies and trailers, and parts	0.027	0.005	0.021	0.036	MotrVehic
33	Truck transportation, warehousing and support activities	0.025	0.008	0.016	0.037	TruckTrans
34	Computer systems design, data proc., other info. srvcs	0.025	0.010	0.000	0.035	InfoTech
35	Electrical equipment, appliances, and components	0.025	0.003	0.021	0.030	Appliance
36	Health	0.022	0.005	0.011	0.028	Health
37	Air transportation	0.021	0.004	0.012	0.029	AirTrans
38	Support activities for mining	0.020	0.013	0.006	0.047	MineSupt
39	Computer and electronic products	0.019	0.004	0.016	0.033	Computer
40	Insurance carriers and related activities	0.018	0.003	0.014	0.027	Insurance
41	Rail transportation	0.016	0.003	0.011	0.022	Rail
42	Miscellaneous manufacturing	0.016	0.004	0.010	0.021	MscManuf
43	Other transportation equipment	0.015	0.002	0.011	0.024	TrnsEq
44	Amusements, gambling, recreation, performing arts	0.013	0.002	0.010	0.017	Amusmt
45	Educational services	0.010	0.003	0.006	0.016	Educ
46	Accommodation	0.010	0.001	0.008	0.012	Hotel
47	Water transportation	0.006	0.001	0.005	0.008	WtrTr
48	Transit and ground passenger transportation	0.004	0.001	0.003	0.006	GrndTr
49	Social assistance	0.004	0.001	0.003	0.006	Soci

Table 1: feedback-strength summary statistics

5.2 Table 2: determinants of feedback strength

Read:

- Centrality of industry is positively related to feedback strength.
- Industry size measures, including the log count of firms and industry market share, also relate positively to FS.
- The industry Herfindahl index relates negatively in some specifications, suggesting concentrated industries are not necessarily more loop-connected.
- In multivariate specifications, centrality and log count of firms are the most important predictors.

Economic message: Large and central industries tend to have stronger feedback loops, but these characteristics do not fully explain the return-predictability evidence.

Table 2
Determinants of industry feedback strength.
This table reports results of panel regressions of industry feedback strength on contemporaneous industry characteristics. Industry centrality is computed following Ahern (2013). Industry market share is the proportion of market capitalization of firms in an industry relative to the market value of all industries. All regressions include month fixed effects. The t-statistics are based on standard errors clustered by industry and are shown in square brackets. The sample, described in Appendix A, covers January 1973 through December 2016.

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Centrality of industry	0.462						0.343	0.357
	[2.46]						[1.87]	[1.92]
Log count of firms in industry		0.008					0.004	0.005
		[3.84]					[1.96]	[2.25]
Log average firm size			0.003				0.001	
			[1.74]				[0.42]	
Industry market share				0.190			0.030	
				[3.58]			[0.42]	
Log age of firms in industry					0.002		0.002	
					[0.72]		[0.87]	
Industry Hirfindahl index						0.020	0.007	
						[2.69]	[1.17]	
R^2	0.23	0.19	0.08	0.11	0.06	0.11	0.27	0.26

Table 3
Momentum profits of industry portfolios sorted on feedback strength.
This table reports raw and factor-adjusted monthly returns of industry momentum portfolios conditional on feedback strength (FS). At the end of each month $t-1$, industries are sorted into terciles by their FS, computed as in Section 2.1. Within each tercile, they are sorted into quintiles by their cumulative intermediate-horizon returns ($t-11$ through $t-7$, inclusive) or recent-horizon returns ($t-6$ through $t-2$, inclusive). The resultant winner-minus-loser value-weighted portfolios are held during month t. Reported are excess returns and alphas from the CAPM, three-factor, and four-factor models, all in percent per month, as well as the associated t-statistics in square brackets. The sample, described in Appendix A, covers January 1973 through December 2016.

Performance measure	Profits (% monthly) of momentum strategies based on industry returns over							
	Intermediate horizon (past 11 to 7 mo)				Recent horizon (past 6 to 2 mo)			
	Low FS	Med FS	High FS	High-low	Low FS	Med FS	High FS	High-low
Excess return	0.38	0.45	0.95	0.57	0.11	0.25	−0.20	−0.32
	[1.66]	[2.26]	[4.10]	[1.74]	[0.46]	[1.13]	[−0.82]	[−0.90]
CAPM alpha	0.39	0.42	0.94	0.55	0.16	0.31	−0.14	−0.31
	[1.66]	[2.07]	[4.00]	[2.09]	[0.65]	[1.41]	[−0.58]	[−1.10]
3-factor alpha	0.49	0.62	1.02	0.53	0.26	0.35	−0.10	−0.36
	[2.14]	[3.14]	[4.34]	[2.00]	[1.02]	[1.59]	[−0.41]	[−1.27]
4-factor alpha	0.07	0.31	0.55	0.49	−0.37	−0.18	−0.74	−0.36
	[0.31]	[1.66]	[2.54]	[1.79]	[−1.73]	[−0.95]	[−3.40]	[−1.27]

5.3 Table 3: baseline echo profits by feedback strength

Read:

- Intermediate-horizon momentum profits rise monotonically with FS.
- Raw intermediate-horizon echo returns are 0.38% per month in low-FS industries, 0.45% in medium-FS industries, and 0.95% in high-FS industries.
- The high-minus-low difference is 0.57% per month in raw returns.
- Three-factor alphas are 0.49%, 0.62%, and 1.02% for low-, medium-, and high-FS industries.
- Recent-horizon momentum profits do not display the same positive pattern; high-FS industries have negative recent-horizon performance.

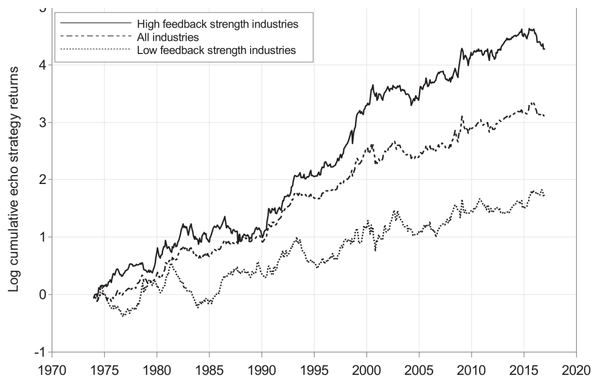
Economic message: Feedback strength explains the intermediate-horizon echo, not generic short-term momentum. This is the central return result.

5.4 Figure 2 and Figure 3: cumulative returns and term structure

Read:

- Figure 2 shows that the high-FS echo strategy outperforms both the unconditional echo strategy and the low-FS echo strategy over the full sample.
- Figure 3 shows the term structure of winner-minus-loser returns across return lags.
- High-FS industries show a wave pattern: weak recent profitability, strong intermediate-lag profitability, and then fading returns.
- Low-FS industries do not display a strong echo-like wave.

Economic message: The figures visualize the paper's main mechanism: feedback loops create delayed return autocorrelation at the intermediate horizon.



ulative returns of industry echo strategies.
plots log cumulative returns of echo strategies that invest in high-FS industries, low-FS industries, or all industries without conditioning
of each month $t-1$, industries are sorted into terciles by their FS, computed as in Section 2.1. Within each tercile, they are sorted into qui
nulative intermediate-horizon returns ($t-11$ through $t-7$, inclusive). The resultant winner-minus-loser value-weighted portfolio is held d

Figure 2: cumulative returns of echo strategies

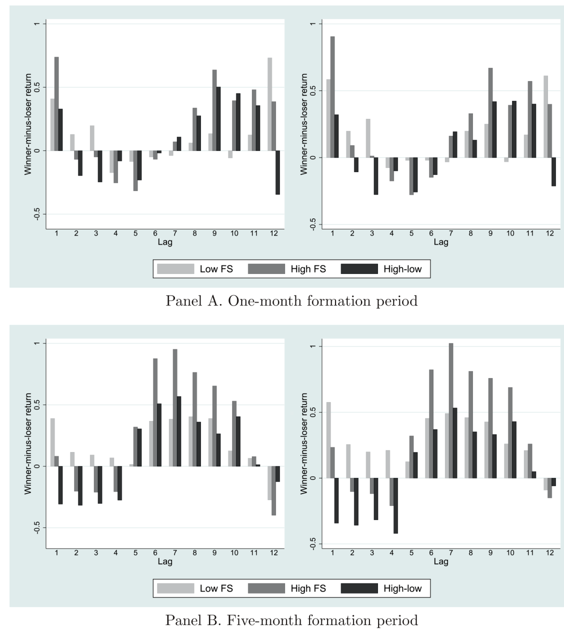


Figure 3: feedback strength and term structure of momentum

5.5 Table 4: conditional echo strategy

Read:

- Table 4 conditions the echo strategy on the feedback strength of winner industries during the formation period.
- When winner-portfolio formation-period FS is in the top quartile, high-FS intermediate-horizon profits are 0.88% per month for tercile sorts and 1.45% for decile sorts.
- When formation-period FS is in the top decile, high-FS intermediate-horizon profits are 1.22% and 1.30% per month depending on the sort.
- Recent-horizon returns remain weak or negative.

Economic message: Echo profits are strongest exactly when the feedback loops of the industries being traded are strong at the time the strategy is formed.

Table 4
Momentum profits of industry portfolios sorted on feedback strength: conditioning on the winner portfolio formation-period feedback strength.

This table reports average returns, in percent per month, of industry momentum portfolios conditional on feedback strength (FS). At the end of each month $t-1$, industries are sorted into terciles by their FS, computed as in Section 2.1. Within each tercile, they are sorted into quintiles by their cumulative intermediate-horizon returns ($t-11$ through $t-7$, inclusive) or recent-horizon returns ($t-6$ through $t-2$, inclusive). The resultant winner-minus-loser value-weighted portfolio is held during month t. The t-statistics are shown in square brackets. The sample, described in Appendix A, covers January 1973 through December 2016.

	Profits (% monthly) of momentum strategies based on industry returns over							
	Intermediate horizon (past 11 to 7 mo)				Recent horizon (past 6 to 2 mo)			
	Low FS	Med FS	High FS	High-low	Low FS	Med FS	High FS	High-low
A. Winner portfolio formation-period FS *in the top quartile of its time-series distribution*								
Winners and losers based on	0.28	0.34	0.88	0.60	−0.27	0.29	−0.57	−0.30
sorts into runup terciles	[1.07]	[2.00]	[3.70]	[1.68]	[−1.03]	[1.43]	[−1.89]	[−0.75]
Winners and losers based on	0.09	0.74	1.45	1.35	−0.10	0.40	−0.47	−0.37
sorts into runup deciles	[0.20]	[2.65]	[2.85]	[1.97]	[−0.18]	[1.25]	[−0.84]	[−0.48]
Number of months	126	251	126	252	127	254	127	254
B. Winner portfolio formation-period FS *in the top decile of its time-series distribution*								
Winners and losers based on	−0.03	0.43	1.22	1.25	−0.32	0.07	−0.85	−0.53
sorts into runup terciles	[−0.08]	[3.09]	[3.35]	[2.36]	[−0.67]	[0.41]	[−1.96]	[−0.82]
Winners and losers based on	−0.73	0.87	1.30	2.03	0.80	−0.020	−0.07	−0.87
sorts into runup deciles	[−0.94]	[3.71]	[1.51]	[1.75]	[0.86]	[−0.07]	[−0.12]	[−0.79]
Number of months	50	403	50	100	51	406	51	102

Table 5
Industry momentum profits and feedback strength: role of limited information.

This table reports the results of panel regressions of industry factor-adjusted returns in month t on its return in month $t-1$; average intermediate-horizon returns (IR, $t-11$ through $t-7$, inclusive); average recent-horizon returns (RR, $t-6$ through $t-2$, inclusive); feedback strength, computed as in Section 2.1; analyst coverage along the feedback loop; and cross terms. For each industry, analyst coverage is defined as the geometric average of the number of analysts covering at least one firm in each two successive industries that form the strongest feedback loop for that industry, normalized by the total number of firms in the departure industry. All returns are factor adjusted, computed as the sum of the intercept and residuals from the in-sample regression of industry excess returns on the three factors of Fama and French (1993). All regressions include month fixed effects. The t-statistics, shown in square brackets, are based on standard errors clustered by time. The sample is described in Appendix A. The availability of analyst coverage data restricts the sample to the period from January 1984 to December 2016.

	Slope coefficients and [t-statistics] from regressions where the dependent variable is industry 3-factor alpha				
Independent variable	(1)	(2)	(3)	(4)	(5)
Last month return	0.005	0.007	0.004	0.005	0.007
	[0.27]	[0.38]	[0.26]	[0.27]	[0.38]
Intermediate-horizon return, IR	−0.001		−0.083	−0.009	
	[−0.02]		[−1.59]	[−0.19]	
Recent-horizon return, RR		0.024			0.006
		[0.49]			[0.08]
Feedback strength, FS	0.001	0.002	0.003	0.005	0.006
	[0.084]	[0.14]	[0.15]	[0.27]	[0.34]
FS × IR	1.994		3.543	1.871	
	[2.93]		[3.24]	[2.70]	
FS × RR		−0.624			−0.029
		[−0.90]			[−0.03]
Analyst coverage			0.001	0.001	0.001
			[0.85]	[0.77]	[0.73]
Analyst coverage × FS			−0.009	−0.010	−0.011
			[−0.44]	[−0.47]	[−0.53]
Analyst coverage × IR			0.186	0.033	
			[1.89]	[0.42]	
Analyst coverage × RR					0.028
					[0.26]
Analyst coverage × FS × IR			−2.754		
			[−2.11]		
Analyst coverage × FS × RR					−0.849
					[−0.66]

5.6 Table 5: analyst coverage and limited information

Read:

- The coefficient on FS × IR is positive and significant, confirming the baseline relation in a regression setting.
- The triple interaction *Analyst coverage* × FS × IR is negative and statistically significant.
- This means the relation between feedback strength and echo profits is weaker when analyst coverage along the loop is high.
- Recent-horizon interactions do not generate the same strong effect.

Economic message: The echo effect is an information-diffusion phenomenon. When more analysts cover linked industries, information travels faster and the delayed echo is weaker.

5.7 Table 6: tracing the feedback channel

Read:

- The table tests whether returns of industries located on the strongest feedback loop help explain the echo.
- The interaction FS × IR × FLP is positive and statistically significant when FLP contains industries in the strongest loop.
- In placebo portfolios based on influenced industries outside the strongest loops, the corresponding interaction is negative and insignificant.
- Standard errors are clustered by month or by month and industry.

Economic message: The echo is not simply driven by broadly related industries. It specifically depends on returns of industries along the actual feedback loop.

Table 6
Industry momentum and feedback strength: investigating the feedback channel.
This table reports the results of panel regressions of industry returns in month t on its return in month $t-1$; average intermediate-horizon returns (IR, $t-11$ through $t-7$, inclusive); average recent-horizon returns (RR, $t-6$ through $t-2$, inclusive); feedback strength, computed as in Section 2.1; feedback loop portfolio (FLP) return; and cross terms. For industry i, FLP consists of up to three industries that are located on the strongest 20% of i's feedback loops and are most influenced by i as measured by $\tilde{w}_{ij}$ from Section 2.1. Industries are sorted by their FLP returns over $t-4$ to $t-1$, inclusive, and the FLP return variable is a dummy equal to one (zero) if an industry's FLP return falls in the top (bottom) decile cross-sectionally. Columns (3) and (4) report results of a placebo test where the industries in the FLP are chosen to be influenced by i as strongly as in specifications (1) and (2) but do not belong to i's strongest loops. All regressions include month fixed effects. The t-statistics, shown in square brackets, are based on standard errors clustered by time or by time and industry. The sample, described in Appendix A, covers January 1973 through December 2016.

	FLP based on industries in the strongest loops		FLP based on industries not in the strongest loops	
Independent variable	(1)	(2)	(3)	(4)
Last month return	0.045	0.045	0.045	0.045
	[2.65]	[2.57]	[2.67]	[2.58]
Intermediate-horizon return, IR	0.003	0.003	0.001	0.001
	[1.25]	[1.48]	[0.44]	[0.54]
Feedback loop portfolio (FLP) return	0.004	0.004	0.002	0.002
	[2.04]	[2.37]	[0.69]	[0.84]
Feedback strength, FS	−0.004	−0.004	−0.015	−0.015
	[−0.27]	[−0.31]	[−1.09]	[−1.34]
FS × IR	0.052	0.052	0.089	0.089
	[1.30]	[1.80]	[2.28]	[3.35]
IR × FLP return	−0.018	−0.018	0.006	0.006
	[−2.66]	[−3.10]	[0.76]	[0.87]
FS × FLP return	−0.083	−0.083	−0.016	−0.016
	[−2.34]	[−2.87]	[−0.36]	[−0.44]
FS × IR × FLP return	0.245	0.245	−0.187	−0.187
	[2.39]	[3.28]	[−1.24]	[−1.54]
Cluster	Month	Month,Ind	Month	Month,Ind

Table 7
Momentum profits of industry portfolios sorted on feedback strength: robustness.
This table reports raw and factor-adjusted monthly returns of industry momentum portfolios conditional on different alternative measures of feedback strength, FS. At the end of each month $t-1$, industries are sorted into terciles by their FS. Within each tercile, they are sorted into quintiles by their cumulative intermediate-horizon returns ($t-11$ through $t-7$, inclusive) or recent-horizon returns ($t-6$ through $t-2$, inclusive). The resultant winner-minus-loser value-weighted portfolios are held during month t. In Panel A, FS is computed based on the PageRank algorithm as described in Section 4.1.1. In Panel B, instead of using the average of the customer-based (w_{ij}^C) and supplier-based (w_{ij}^S) influence measures as in Section 2.1 to compute FS, the maximum of the two influence measures is used. In Panel C, FS is computed for each industry as the average of the feedback strength associated with its ten, rather than one, strongest feedback loops. Reported are excess returns and alphas from the CAPM, three-factor, and four-factor models, all in percent per month, as well as the associated t-statistics in square brackets. The sample, described in Appendix A, covers January 1973 through December 2016.

Performance measure	Profits (% monthly) of momentum strategies based on industry returns over							
	Intermediate horizon (past 11 to 7 mo)				Recent horizon (past 6 to 2 mo)			
	Low FS	Med FS	High FS	High-low	Low FS	Med FS	High FS	High-low
A. Feedback strength based on the PageRank algorithm								
Excess return	0.21	0.38	1.10	0.88	0.42	−0.13	−0.25	−0.67
	[0.92]	[2.01]	[4.45]	[2.60]	[1.62]	[−0.60]	[−1.00]	[−1.87]
CAPM alpha	0.23	0.40	1.04	0.81	0.49	−0.05	−0.23	−0.72
	[0.98]	[2.14]	[4.19]	[2.88]	[1.86]	[−0.25]	[−0.94]	[−2.55]
3-factor alpha	0.33	0.52	1.18	0.85	0.59	−0.04	−0.17	−0.76
	[1.39]	[2.75]	[4.73]	[3.01]	[2.24]	[−0.20]	[−0.68]	[−2.66]
4-factor alpha	−0.16	0.16	0.79	0.96	−0.06	−0.57	−0.78	−0.72
	[−0.77]	[0.92]	[3.31]	[3.31]	[−0.26]	[−2.97]	[−3.50]	[−2.46]
B. Feedback strength based on the maximum influence of the supplier or customer								
Excess return	0.34	0.53	1.03	0.69	0.29	−0.10	−0.12	−0.41
	[1.44]	[2.80]	[4.04]	[2.00]	[1.18]	[−0.47]	[−0.45]	[−1.15]
CAPM alpha	0.35	0.55	1.00	0.65	0.35	0.01	−0.11	−0.46
	[1.47]	[2.88]	[3.90]	[2.25]	[1.40]	[0.03]	[−0.42]	[−1.61]
3-factor alpha	0.42	0.66	1.15	0.73	0.42	0.07	−0.08	−0.51
	[1.82]	[3.53]	[4.48]	[2.50]	[1.67]	[0.31]	[−0.32]	[−1.75]
4-factor alpha	−0.01	0.32	0.68	0.68	−0.20	−0.41	−0.74	−0.54
	[−0.03]	[1.81]	[2.82]	[2.31]	[−0.90]	[−2.08]	[−3.27]	[−1.83]
C. Feedback strength based on multiple feedback loops								
Excess return	0.28	0.39	0.93	0.65	0.24	0.35	−0.43	−0.67
	[1.20]	[1.96]	[4.06]	[1.99]	[0.94]	[1.54]	[−1.69]	[−1.86]
CAPM alpha	0.29	0.34	0.95	0.66	0.30	0.41	−0.34	−0.64
	[1.23]	[1.69]	[4.13]	[2.48]	[1.18]	[1.82]	[−1.36]	[−2.29]
3-factor alpha	0.38	0.47	1.02	0.63	0.40	0.48	−0.33	−0.73
	[1.65]	[2.38]	[4.36]	[2.36]	[1.57]	[2.07]	[−1.30]	[−2.56]
4-factor alpha	−0.01	0.12	0.53	0.54	−0.21	−0.12	−0.95	−0.74
	[−0.05]	[0.63]	[2.49]	[1.98]	[−0.95]	[−0.63]	[−4.28]	[−2.55]

5.8 Table 7: alternative feedback-strength measures

Read:

- Panel A computes FS using a PageRank-style algorithm.
- Panel B uses the maximum of the customer-based and supplier-based influence measures.

- Panel C uses multiple feedback loops rather than only the single strongest loop.
- Across these alternative definitions, intermediate-horizon profits are still larger for high-FS industries.
- Recent-horizon results remain weak or negative.

Economic message: The result is not an artifact of one graph-theoretic implementation. Different reasonable measures of feedback strength produce similar conclusions.

5.9 Table 8: controlling for centrality and number of firms

Read:

- The regression controls for industry centrality and log firm count, along with interactions with IR and RR.
- The FS × IR coefficient remains positive and statistically significant.
- Centrality interactions are not the main driver of the echo strategy returns.
- Log firm count does not eliminate the FS × IR effect.

Economic message: Feedback strength matters beyond general industry centrality and industry size. The relevant object is the loop structure of the trade network.

Table 8
Industry momentum profits and feedback strength: robustness to controlling for industry centrality and size.

This table reports the results of panel regressions of industry factor-adjusted returns in month t on its return in month $t-1$; average intermediate-horizon returns (IR, $t-11$ through $t-7$, inclusive); average recent-horizon returns (RR, $t-6$ through $t-2$, inclusive); feedback strength, computed as in Section 2.1; industry centrality, computed following Ahern (2013); log of the count of firms in the industry; and cross terms. All returns are factor adjusted, computed as the sum of the intercept and residuals from the in-sample regression of industry excess returns on the three factors of Fama and French (1993). All regressions include month fixed effects. The t-statistics, shown in square brackets, are based on standard errors clustered by time. The sample, described in Appendix A, covers January 1973 through December 2016.

	Slope coefficients and [t-statistics] from regressions where the dependent variable is industry 3-factor alpha				
Independent variable	(1)	(2)	(3)	(4)	(5)
Last month return	0.031	0.031	0.034	0.031	0.034
	[2.05]	[2.05]	[2.22]	[2.03]	[2.21]
Intermediate-horizon return, IR	0.004	−0.004		−0.096	
	[0.12]	[−0.10]		[−0.95]	
Recent-horizon return, RR			0.014		−0.164
			[0.32]		[−1.52]
Feedback strength, FS	0.000	0.001	0.002	−0.001	0.001
	[−0.01]	[0.08]	[0.17]	[−0.06]	[0.05]
FS × IR	2.212	2.097		2.106	
	[3.76]	[3.20]		[3.53]	
FS × RR			−0.576		−0.664
			[−0.82]		[−1.01]
Centrality		−0.006	−0.007		
		[−0.48]	[−0.52]		
Centrality × IR		0.701			
		[0.61]			
Centrality × RR			0.754		
			[0.64]		
Log firm count				0.000	0.000
				[1.20]	[1.26]
Log firm count × IR				0.026	
				[1.04]	
Log firm count × RR					0.048
					[1.76]

6 Short summary

- The paper explains the momentum echo documented by Novy-Marx (2012).
- Its main idea is that industry shocks move through trade networks and can later feed back to the original industry.
- The authors construct a feedback-strength measure using BEA input-output data and graph-theoretic shortest-path methods.
- Industries with high feedback strength have much larger intermediate-horizon momentum profits.
- The relation is not present for recent-horizon momentum profits.
- The term structure of momentum in high-FS industries has a clear wave-like echo pattern.
- Analyst coverage along the feedback loop weakens the relation, consistent with limited-information models.
- Returns of industries on the strongest feedback loop help explain the echo; placebo loop portfolios do not.

- The findings are robust to PageRank-based feedback strength, maximum customer/supplier influence, multiple-loop measures, and controls for centrality and industry size.
- The paper identifies intersectoral trade networks as a mechanism that determines not only where shocks propagate but also when return predictability appears.

7 Conclusion

7.1 Core conclusion

Sharifkhani and Simutin show that feedback loops in industry trade networks explain the term structure of momentum profits. The momentum echo is strongest in industries whose customer-supplier network links provide a strong route for shocks to travel away from the industry and later return to it.

7.2 Economic intuition

A shock to an industry affects its customers and suppliers. If information diffuses slowly across segmented industries, prices of linked industries adjust with delay. When the trade network contains a loop back to the original industry, the shock can return to the source industry after several months. This generates intermediate-term autocorrelation and explains why momentum profits peak at the echo horizon.

7.3 Takeaway

The paper connects asset pricing momentum to real economic networks. Momentum is not only a behavioral or statistical phenomenon; in this setting, it reflects delayed propagation of industry shocks through production and trade links. The loop structure of the input-output network determines the timing and profitability of industry momentum.